

# Data Dictionary

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## Accessing the Blob Containers on Microsoft Azure Storage Explorer

### [CheXpert: Chest X-Rays]

To use the CheXpert files, request access using this link:

<https://stanfordaimi.azurewebsites.net/datasets/8cbd9ed4-2eb9-4565-affc-111cf4f7ebe2>

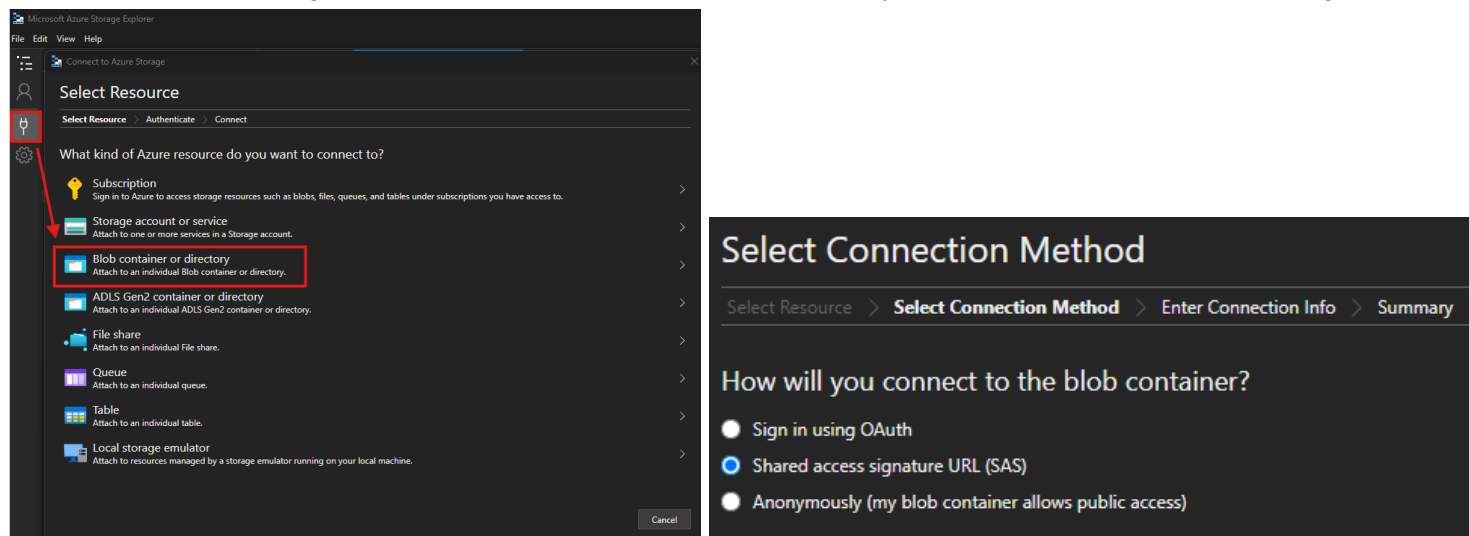
### [CheXlocalize]

To access the test radiologist-labeled datasets, request access using this link:

<https://stanfordaimi.azurewebsites.net/datasets/23c56a0d-15de-405b-87c8-99c30138950c>

For each, a link to the blob will be provided which can be accessed using some Microsoft Azure application. We used Microsoft Azure Storage Explorer which provides a useful UI for navigation purposes.

Open the connect dialog box, then select “Blob container or directory.” Then, select “Shared access signature URL (SAS)”



On the following page, paste the link to the blob in the “Blob container or directory SAS URL” field, and the “Display name” field will auto populate. Then, click Next. If everything was done correctly, the files should appear in the Storage Explorer UI. Only the below files and folders are necessary for this project

## Explanation of Files From the CheXpert Blob

| Name                                       | Content Type          | Size      | Last Modified | Parent Directory                           | Description                                                                                                        | Required      |
|--------------------------------------------|-----------------------|-----------|---------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------|
| CheXpert-v1.0 batch 1 (validate & csv).zip | Compressed Zip Folder | 486.04 MB | 12/30/2023    | -                                          | Contains the <b>radiologist-labeled validation</b> images (in /valid/) as well as associated metadata in valid.csv | <b>YES **</b> |
| valid.csv                                  | CSV                   | 31 KB     | 1/20/2019     | CheXpert-v1.0 batch 1 (validate & csv).zip | Contains the <b>radiologist-labeled validation</b> image paths and associated metadata                             | <b>YES **</b> |
| CheXpert-v1.0 batch 2 (train 1).zip        | Compressed Zip Folder | 162.39 GB | 12/30/2023    | -                                          | Contains the first set of training images                                                                          | <b>YES</b>    |
| CheXpert-v1.0 batch 3 (train 2).zip        | Compressed Zip Folder | 184.82 GB | 12/30/2023    | -                                          | Contains the second set of training images                                                                         | <b>YES</b>    |
| CheXpert-v1.0 batch 4 (train 3).zip        | Compressed Zip Folder | 91.09 GB  | 12/30/2023    | -                                          | Contains the third set of training images                                                                          | <b>YES</b>    |
| README.md                                  | Plain-text            | 3.21 KB   | 8/9/2021      | -                                          | A plain-text file which documents the dataset, the images, and the labeling                                        | <b>YES *</b>  |
| train_chexbert.csv                         | CSV                   | 22.06 MB  | 8/9/2021      | -                                          | The training labels produced by the CheXbert labeler                                                               | <b>YES</b>    |

\* The README file is not required to run the project, but it contains useful information about the datasets.

\*\* The radiologist-labeled validation files are used to benchmark our model's performance against human-labeling, but are not required to train and run the model.

## Explanation of Files From the CheXlocalize Blob

| Name            | Content Type | Size    | Last Modified | Parent Directory                           | Description                                                                            | Required      |
|-----------------|--------------|---------|---------------|--------------------------------------------|----------------------------------------------------------------------------------------|---------------|
| test            | Folder       | 1.94 MB | 8/9/2021      | -                                          | Contains limited demographic information for each patient in the dataset               | <b>YES **</b> |
| test_labels.csv | CSV          | 31 KB   | 1/20/2019     | CheXpert-v1.0 batch 1 (validate & csv).zip | Contains the <b>radiologist-labeled validation</b> image paths and associated metadata | <b>YES **</b> |

\*\* The radiologist-labeled test files are used to benchmark our model's performance against human-labeling, but are not required to train and run the model.

## Starting Schema Configuration

In order for images to be located properly, the source data must be downloaded from the blob and organized into the following directory structure:

```
| MAIN_DIR
|----| raw_data
|-----| CheXpert-v1.0 batch 1 (validate & csv)
|-----| valid
|-----| valid.csv
|-----| CheXpert-v1.0 batch 2 (train 1)
|-----| CheXpert-v1.0 batch 3 (train 2)
|-----| CheXpert-v1.0 batch 4 (train 3)
|-----| test
|-----| test_labels.csv
|-----| train_cheXbert.csv
```

### Directories

**MAIN\_DIR**: The directory containing all project files

**raw\_data**: The directory containing all images and data frames as downloaded from the blobs

### Patient directory structure

```
| Patient ID
|----| Study ID
|-----| Frontal View / Lateral View
```

### Example

**File path:** **raw\_data**/CheXpert-v1.0 batch 3 (train 2)/patient21521/study3/view1\_frontal.jpg

## **Required Non-Standard Python Libraries**

| Library Name        | Documentation Link                                                                                                      | Description                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| torch               | <a href="https://pytorch.org/get-started/locally/">https://pytorch.org/get-started/locally/</a>                         | This library powers our neural network models                                                           |
| torchvision         | <a href="https://pytorch.org/get-started/locally/">https://pytorch.org/get-started/locally/</a>                         | Torchvision contains the pre-trained ResNet50 and DenseNet121 models for transfer learning              |
| numpy-hilbert-curve | <a href="https://github.com/PrincetonLIPS/numpy-hilbert-curve">https://github.com/PrincetonLIPS/numpy-hilbert-curve</a> | We use this library during our Grad-CAM analysis to better encode spatial relationships in a 1-D vector |
| grad-cam            | <a href="https://github.com/jacobgil/pytorch-grad-cam">https://github.com/jacobgil/pytorch-grad-cam</a>                 | Grad-CAM is used to visualize the decision-making of our model                                          |

## Data Transformation Steps

If the image files and label data frames are saved according to pages 2-4, then the below actions can be performed using Preprocessing1\_EnvironmentSetup.ipynb (Steps A-B) and Preprocessing2\_ImagePreprocessing.ipynb (Step C)

| Step                                                                                                                                              | Description                                                                                                                                                                                   | Input Signature | Relevant Prior Step | Output Signature | Tool/Library Required                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|------------------|------------------------------------------------------|
| <b>(A) File Organization</b>                                                                                                                      |                                                                                                                                                                                               |                 |                     |                  |                                                      |
| A1                                                                                                                                                | Download, extract, and organize the required files into the required directory structure                                                                                                      | -               | Reference pages 2-4 | -                | File Explorer, Microsoft Azure Storage Explorer      |
| A2                                                                                                                                                | Copy the directory structures of the image folders. This is performed in Preprocessing1_EnvironmentSetup                                                                                      | -               | A1                  | -                | File Explorer, Preprocessing1_EnvironmentSetup.ipynb |
| <b>(B) Data Frame Preparation</b>                                                                                                                 |                                                                                                                                                                                               |                 |                     |                  |                                                      |
| Though the exact processing steps are detailed below, the code to run these functions are contained within Preprocessing1_EnvironmentSetup.ipynb. |                                                                                                                                                                                               |                 |                     |                  |                                                      |
| B1                                                                                                                                                | From train_cheXbert.csv, extract the rows with file paths to images contained within CheXpert-v1.0 batch 4 (train 3) as the test set                                                          | Data Frame      | A2                  | Data Frame       | Pandas                                               |
| B2                                                                                                                                                | From train_cheXbert.csv, extract the rows with file paths to images contained within CheXpert-v1.0 batch 2 (train 1) and CheXpert-v1.0 batch 3 (train 2) as the combined train_validation set | Data Frame      | A2                  | Data Frame       | Pandas                                               |
| B3                                                                                                                                                | Shuffle the train_validation set                                                                                                                                                              | Data Frame      | B2                  | Data Frame       | Pandas                                               |
| B4                                                                                                                                                | Split the train_validation set into a train set and validation set using a 80%-20% ratio                                                                                                      | Data Frame      | B3                  | Data Frame       | Pandas, sklearn.model_selection.train_test_split()   |
| B5                                                                                                                                                | From valid.csv, extract the rows with file paths to images contained within CheXpert-v1.0 batch 1 (validate & csv) as the radiologist-labeled validation set                                  | Data Frame      | A2                  | Data Frame       | Pandas                                               |
| B6                                                                                                                                                | From test_labels.csv, extract the rows with file paths to images contained within test as the radiologist-labeled validation set                                                              | Data Frame      | A2                  | Data Frame       | Pandas                                               |
| B7                                                                                                                                                | Add new columns containing file paths to preprocessed image variations                                                                                                                        | Data Frame      | B1-B6               | Data Frame       | Pandas                                               |
| B8                                                                                                                                                | Filter out unnecessary conditions columns (Keep only Cardiomegaly, Consolidation, Edema, Atelectasis, Pleural Effusion)                                                                       | Data Frame      | B7                  | Data Frame       | Pandas                                               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                   |              |     |              |                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|--------------|---------------------------------------|
| B9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Remove lateral images from the data frames                                                                                                                                                                                                        | Data Frame   | B8  | Data Frame   | Pandas                                |
| B10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Remove rows where all five condition columns are 0                                                                                                                                                                                                | Data Frame   | B9  | Data Frame   | Pandas                                |
| B11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Create train_df_oversampled                                                                                                                                                                                                                       | Data Frame   | B10 | Data Frame   | Pandas,<br>scripts.prepare.resample() |
| B12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sample 30,000 images from the training dataset with weighted replacement                                                                                                                                                                          | Data Frame   | B11 | Data Frame   | Pandas,<br>scripts.prepare.resample() |
| B13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Downsample the original training set by removing 30,000 images randomly                                                                                                                                                                           | Data Frame   | B12 | Data Frame   | Pandas,<br>scripts.prepare.resample() |
| B14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Concatenate the sample of 30,000 with the downsampled training set                                                                                                                                                                                | Data Frame   | B13 | Data Frame   | Pandas                                |
| <p><b>(C) Image Preprocessing Pipelines (Performed up-front for both the raw and enhanced images)</b></p> <p>Though the exact processing steps are detailed below, these are all wrapped into the <b>scripts.preprocessing.run_pipeline()</b> function for convenience. The code to run this function is also contained within Preprocessing2_ImagePreprocessing.ipynb</p> <p><b>scripts.preprocessing.run_pipeline(df, dims, skip_if_image_exists=True, **kwargs)</b></p> <p><b>df</b> [ <i>Pandas Data Frame</i> ]: Data frame containing the image labels and file paths<br/> <b>dims</b> [ <i>List</i> ]: List of output image dimensions. This must be [224], [384] or [284, 384]. If converting to a single dimension, this should still be input as a list.<br/> <b>skip_if_image_exists</b> [ <i>bool</i> ]: If a file path already leads to an existing image, continue to the next file path<br/> <b>**kwargs</b> [ <i>dict</i> ], optional: A dictionary containing keyword arguments from the image processing pipeline. By default, these are the settings:<br/>             <b>usm_weight</b> = 1.2 [ <i>float</i> ]: The weight value for unsharp masking<br/>             <b>usm_sigma</b>= 10 [ <i>float</i> ]: The Gaussian blur parameter for unsharp masking<br/>             <b>he_sigma</b>= 5 [ <i>float</i> ]: The Gaussian blur parameter for histogram equalization<br/>             <b>scale_min</b>= 0 [ <i>float</i> ]: The output minimum pixel value<br/>             <b>scale_max</b>= 255 [ <i>float</i> ]: The output maximum pixel value</p> |                                                                                                                                                                                                                                                   |              |     |              |                                       |
| C1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Run the image processing pipeline on a specified data frame. This function will result in two saved copies (one “raw” and one “enhanced”) to the output file path associated to the input dims<br><br>Below are the step-by-step transformations: | Data Frame   | B14 | .jpeg images | scripts.preprocessing.run_pipeline()  |
| C2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Load image from the source file path as an integer Tensor                                                                                                                                                                                         | .jpeg file   | C1  | Int Tensor   | torchvision.io.decode_image()         |
| C3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Resize the training/validation/testing images to either 224x224 or 384x384 with bilinear interpolation                                                                                                                                            | Int Tensor   | C2  | Float Tensor | v2.Resize()                           |
| C4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Convert float Tensor to numpy array                                                                                                                                                                                                               | Float Tensor | C3  | Float array  | tensor.numpy()                        |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                 |              |     |              |                                                |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----|--------------|------------------------------------------------|
| C5                                                                                           | Rescale the image pixel values to the range [0,255]                                                                                                                                                                                                                                                                                             | Float array  | C4  | Float array  | scripts.preprocessing.scale_range()            |
| C6                                                                                           | Convert float array to uint8 array                                                                                                                                                                                                                                                                                                              | Float array  | C5  | UInt8 array  | arr.astype(np.uint8)                           |
| C7                                                                                           | Extract the 2nd and 3rd axes (img = img[0,:,:]) to get a 2D representation                                                                                                                                                                                                                                                                      | UInt8 array  | C6  | UInt8 array  | arr[0, :, :]                                   |
| C8                                                                                           | (Enhanced Images) Rescale the image pixel values to the range [0,255]                                                                                                                                                                                                                                                                           | UInt8 array  | C7  | Float array  | scripts.preprocessing.scale_range()            |
| C9                                                                                           | (Enhanced Images) Sharpen the image using unsharp masking                                                                                                                                                                                                                                                                                       | Float array  | C8  | Float array  | scripts.preprocessing.unsharp_masking()        |
| C10                                                                                          | (Enhanced Images) Equalize the histogram to increase contrast                                                                                                                                                                                                                                                                                   | Float array  | C9  | Float array  | scripts.preprocessing.histogram_equalization() |
| C11                                                                                          | Convert array to type uint8                                                                                                                                                                                                                                                                                                                     | Float array  | C10 | UInt8 array  | arr.astype(np.uint8)                           |
| C12                                                                                          | Convert integer array to Image object                                                                                                                                                                                                                                                                                                           | UInt8 array  | C11 | Image object | Image.fromarray(arr)                           |
| C13                                                                                          | Save the processed images as .jpeg files with 0.95 quality                                                                                                                                                                                                                                                                                      | Image object | C12 | .jpeg file   | Image.save()                                   |
| <b>(D) Training Image Processing Pipeline (Performed only within the PyTorch dataloader)</b> |                                                                                                                                                                                                                                                                                                                                                 |              |     |              |                                                |
| D1                                                                                           | Load Image as a 3D Tensor (RGB instead of Grayscale)                                                                                                                                                                                                                                                                                            | .jpeg file   | C13 | Int Tensor   | torchvision.io.decode_image()                  |
| D2                                                                                           | Convert integer tensor to float                                                                                                                                                                                                                                                                                                                 | Int Tensor   | D1  | Float Tensor | Tensor.float()                                 |
| D3                                                                                           | Randomly rotate by a 90-degree interval                                                                                                                                                                                                                                                                                                         | Float Tensor | D2  | Float Tensor | torch.rot90()                                  |
| D4                                                                                           | Randomly rotate by a 30 degree interval                                                                                                                                                                                                                                                                                                         | Float Tensor | D3  | Float Tensor | v2.functional.rotate()                         |
| D5                                                                                           | Randomly crop either 0, 16, 32, or 48 pixels from the edge                                                                                                                                                                                                                                                                                      | Float Tensor | D4  | Float Tensor | v2.CenterCrop()                                |
| D6                                                                                           | If a non-zero number of pixels were cropped, resize back to the original dimensions with bilinear interpolation                                                                                                                                                                                                                                 | Float Tensor | D5  | Float Tensor | v2.Resize()                                    |
| D7                                                                                           | Normalize according to the specified mean and standard deviation values specified here:<br><br>ResNet: <a href="https://pytorch.org/hub/pytorch_vision_resnet/">https://pytorch.org/hub/pytorch_vision_resnet/</a><br>DenseNet: <a href="https://pytorch.org/hub/pytorch_vision_densenet/">https://pytorch.org/hub/pytorch_vision_densenet/</a> | Float Tensor | D6  | Float Tensor | v2.Normalize()                                 |



## Explanation of Variables

train\_cheXbert.csv contains the file path to each x-ray image along with corresponding features that indicate the presence (or lack thereof) of 14 pathological conditions. It also contains some limited demographic information regarding the patient as well as the configuration parameters of the x-ray itself.

| Field Name                | Data Type | Field Size | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    | Field Type             |
|---------------------------|-----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Path                      | TEXT      | 58         | The file path leading to the x-ray image.                                                                                                                                                                                                                                                                                                                                                                                                      | ID                     |
| Sex                       | TEXT      | 7          | The gender of the patient.                                                                                                                                                                                                                                                                                                                                                                                                                     | Demographic            |
| Age                       | INT64     | 3          | The age of the patient when the study was performed.                                                                                                                                                                                                                                                                                                                                                                                           | Demographic            |
| Frontal/Lateral           | TEXT      | 7          | This indicates whether the x-ray view was taken from a frontal position, or a lateral position                                                                                                                                                                                                                                                                                                                                                 | X-ray configuration    |
| AP/PA                     | TEXT      | 3          | X-ray code which indicates the view angle<br><br>Posteroanterior (PA) refers to when the x-ray beam passes through the patient from the back to the front<br><br>Anteroposterior (AP) refers to when the x-ray beam passes through the patient from the front to the back<br><br>Lower Lumbar (LL) refers to views of the lumbar spine and sacrum area<br><br>RL refers to views taken from either the right (R) or left (L) sides of the body | X-ray configuration    |
| Enlarged Cardiomedastinum | FLOAT64   | 3          | This indicates if the cavity containing the heart and other structures is enlarged.                                                                                                                                                                                                                                                                                                                                                            | Pathological condition |
| Cardiomegaly              | FLOAT64   | 3          | This indicates the presence of an enlarged heart.                                                                                                                                                                                                                                                                                                                                                                                              | Pathological condition |
| Lung Opacity              | FLOAT64   | 3          | This indicates the presence of hazy, dense areas in the lung that should be darker.                                                                                                                                                                                                                                                                                                                                                            | Pathological condition |
| Lung Lesion               | FLOAT64   | 3          | This indicates an abnormal growth in the lung tissue.                                                                                                                                                                                                                                                                                                                                                                                          | Pathological condition |
| Edema                     | FLOAT64   | 3          | This indicates the presence of fluid collection in the air sacs                                                                                                                                                                                                                                                                                                                                                                                | Pathological condition |
| Consolidation             | FLOAT64   | 3          | This indicates when the air within small airways of the lungs is replaced with a fluid, solid, or other material (such as pus, blood, water, etc.).                                                                                                                                                                                                                                                                                            | Pathological condition |

|                  |         |   |                                                                                                                                                      |                        |
|------------------|---------|---|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Pneumonia        | FLOAT64 | 3 | This indicates the presence of a lung infection causing inflammation and fluid buildup in the lungs.                                                 | Pathological condition |
| Atelectasis      | FLOAT64 | 3 | This indicates the presence of the collapse of a lung (or part of a lung) which occurs when the alveoli lose air and deflate.                        | Pathological condition |
| Pneumothorax     | FLOAT64 | 3 | This indicates the presence of air leakage from the lungs into the space between the chest wall and the lungs. This can cause the lungs to collapse. | Pathological condition |
| Pleural Effusion | FLOAT64 | 3 | This indicates the presence of a collection of fluid in the space between the chest wall and the lungs                                               | Pathological condition |
| Pleural Other    | FLOAT64 | 3 | This indicates the presence of some condition affecting the pleura (The membrane lining the chest wall and lungs)                                    | Pathological condition |
| Fracture         | FLOAT64 | 3 | This indicates the presence of a break in a bone.                                                                                                    | Pathological condition |
| Support Devices  | FLOAT64 | 3 | This indicates the presence of support devices in the image.                                                                                         | Pathological condition |
| No Finding       | FLOAT64 | 3 | This indicates the absence of all above pathologies                                                                                                  | Pathological condition |

## **Breakdown of Pathological Condition Values**

For each training instance and pathological condition, a value indicates the presence of (or lack thereof) the specified condition. These values were extracted using the CheXbert labeler and are provided in the blob:

- 1: The condition is present
- 0: The condition is not present
- -1: The labeler was uncertain if this condition was present
  - For our analysis, we map these values to 2
- N/A: The condition was not mentioned in radiological reports
  - For our analysis, we map these values to 0